

11 **(short)** Two coaxial cylindrical surfaces have radii r_1 and r_2 , $r_2 > r_1$, height h , and $h \gg r_2$. The inner cylindrical surface carries a charge of $+Q$, the outer cylindrical surface carries a charge of $-Q$, and the two cylindrical surfaces rotate along the central axis at a uniform angular velocity ω , as shown in Fig. 6. Ignore fringe effects.

(a) Derive expressions for the current densities on the two cylindrical surfaces. [5]

(b) Derive expressions for the distribution of magnetic flux density $\mathbf{B}$ in the regions inside the central cylinder, between the two cylinders, and outside the outer cylinder. [5]

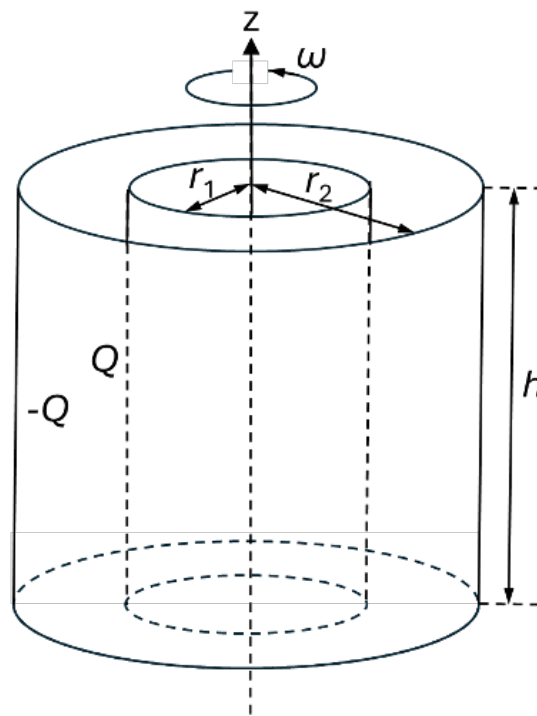


Fig. 6